



MD-100 Windows Client

Module 06 : Configure data access and usage

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Module agenda



Explore Windows client file systems



Configure and manage file access



Configure and manage shared folders

Manage user files

Lesson 1: Explore Windows client file systems



Lesson 1: Configure and manage file access

- Examine the FAT file system
- Examine the NTFS file system
- Examine the ReFS file system

Examine the FAT file system

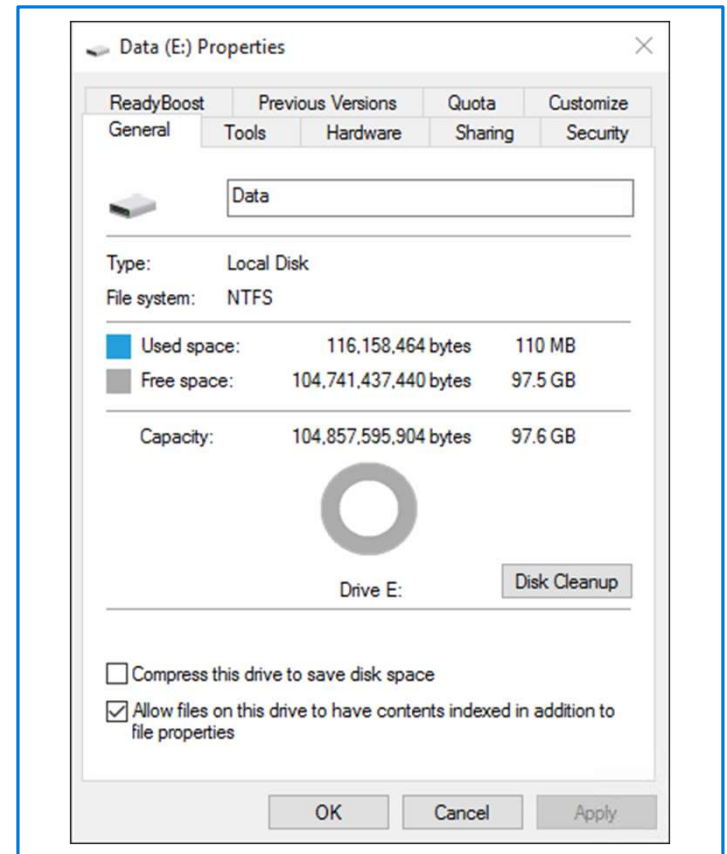
- Supported by most operating systems
- For small volumes with simple folder structure
- Often used for removable media
- Windows supports FAT, FAT32, and exFAT:
 - Maximum volume size, cluster size, and number of files
- No support for security and other NTFS features

Examine the NTFS file system

Default file system in Windows

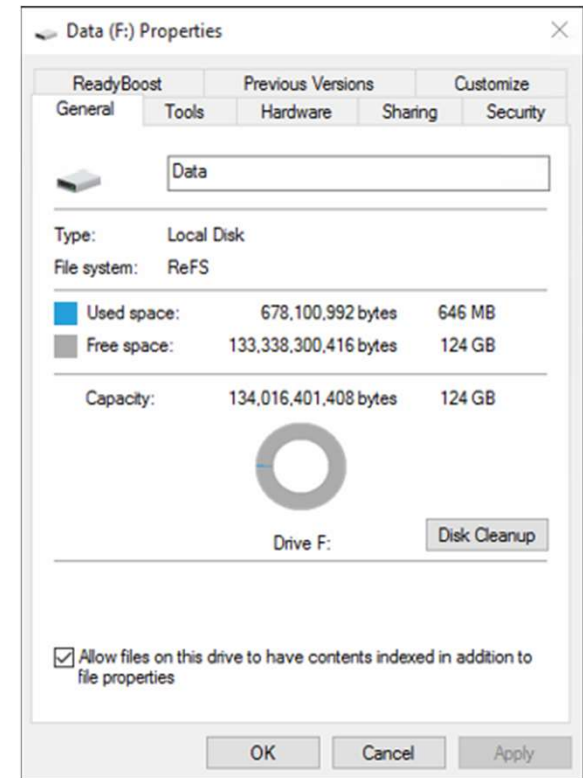
Supports many features not available with FAT:

- Security and auditing
- Large volumes
- Quotas
- Compression
- Encryption
- Reliability
- Other advanced features



Examine the ReFS file system

- Provides the highest level of resiliency and scalability
- Can be used only on mirrored storage spaces:
 - Automatically detects and corrects errors
 - Two-way or three-way mirror
- Supports security and auditing
- Does not support:
 - Compression
 - Quotas
 - Encryption
 - Volume shrinking



Knowledge Check

Lesson 2: Configure and manage file access

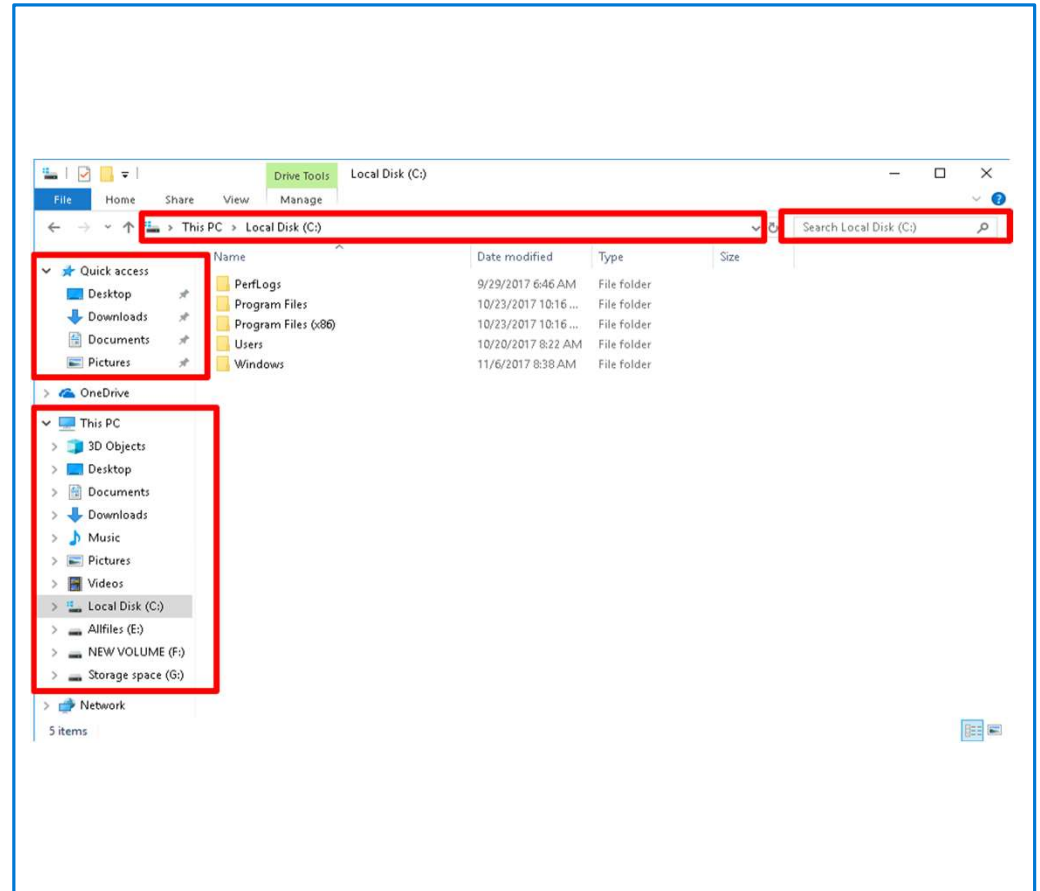


Lesson 2: Configure and manage file access

- Explore tools used for managing files and folders
- Understand file and folder permissions
- Understand permission inheritance
- Implement conditions to limit file and folder access
- Understand effective permissions
- Use the effective access feature
- Examine permissions when copying and moving files

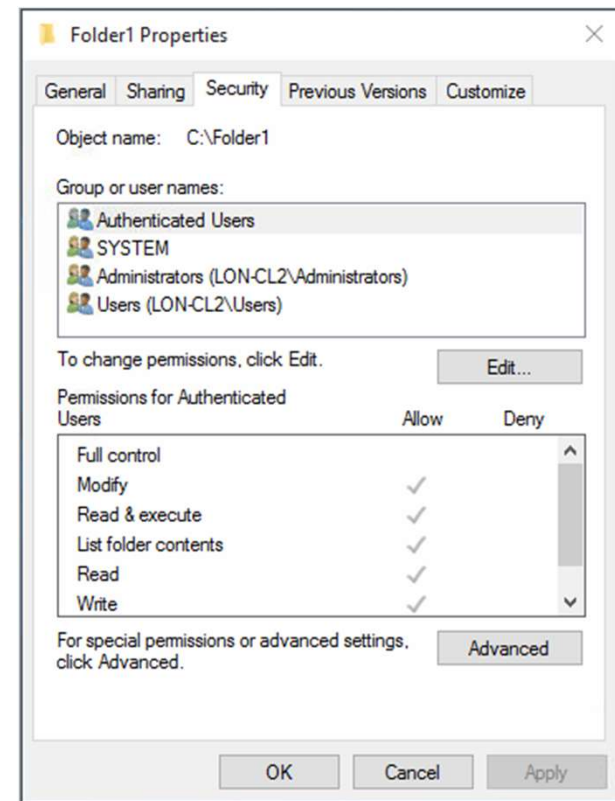
Explore tools used for managing files and folders

- Command prompt:
 - **cd** or **chdir** to change the parent directory
 - **del**, **md**, **move**, and other commands to manage files and folders
 - **icacls** to displays and modifies permissions
- Windows PowerShell:
 - **Set-Location**, **Remove-Item**, and other cmdlets
 - Same aliases as command prompt commands
 - **Set-ACL** to manage file permissions

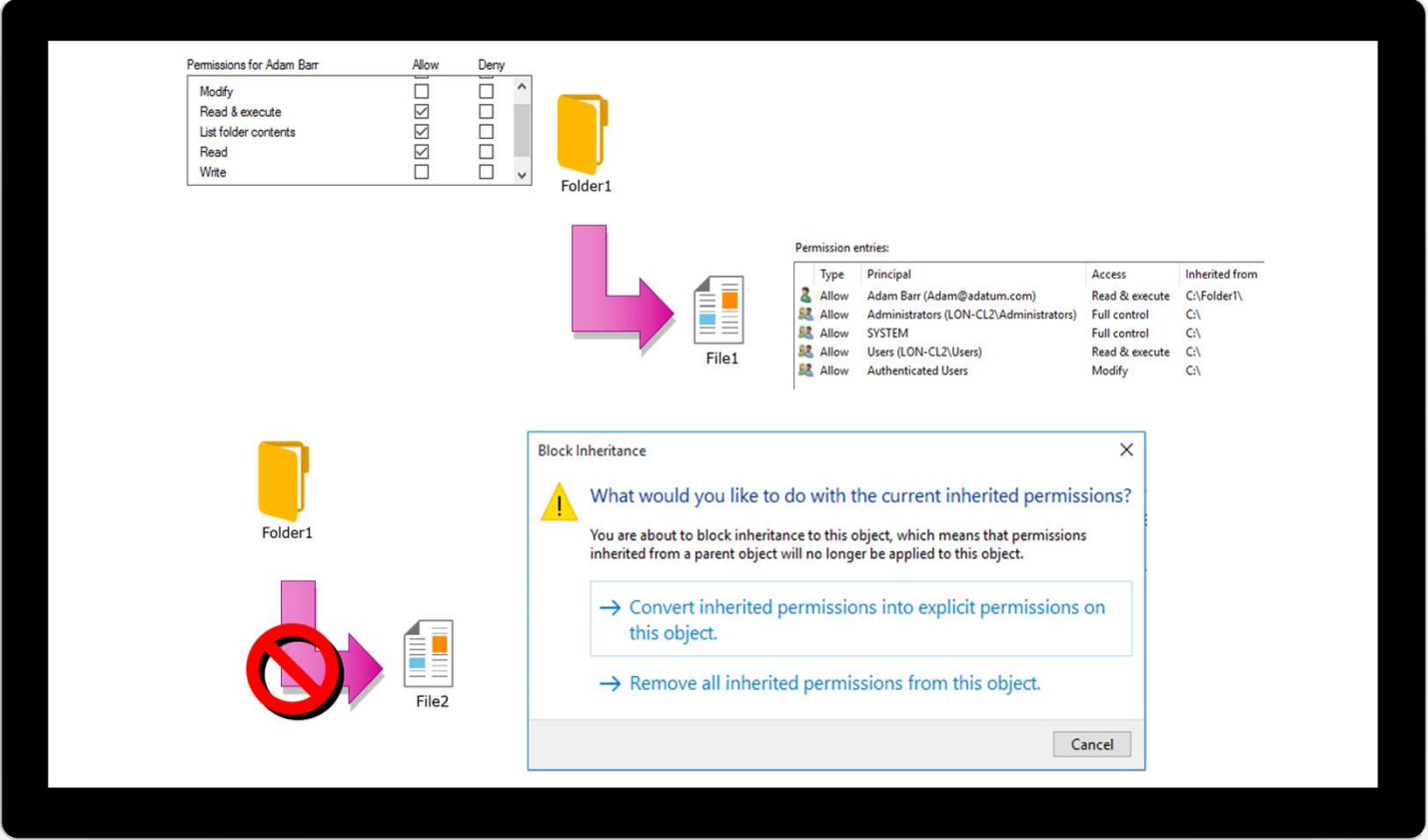


Understand file and folder permissions

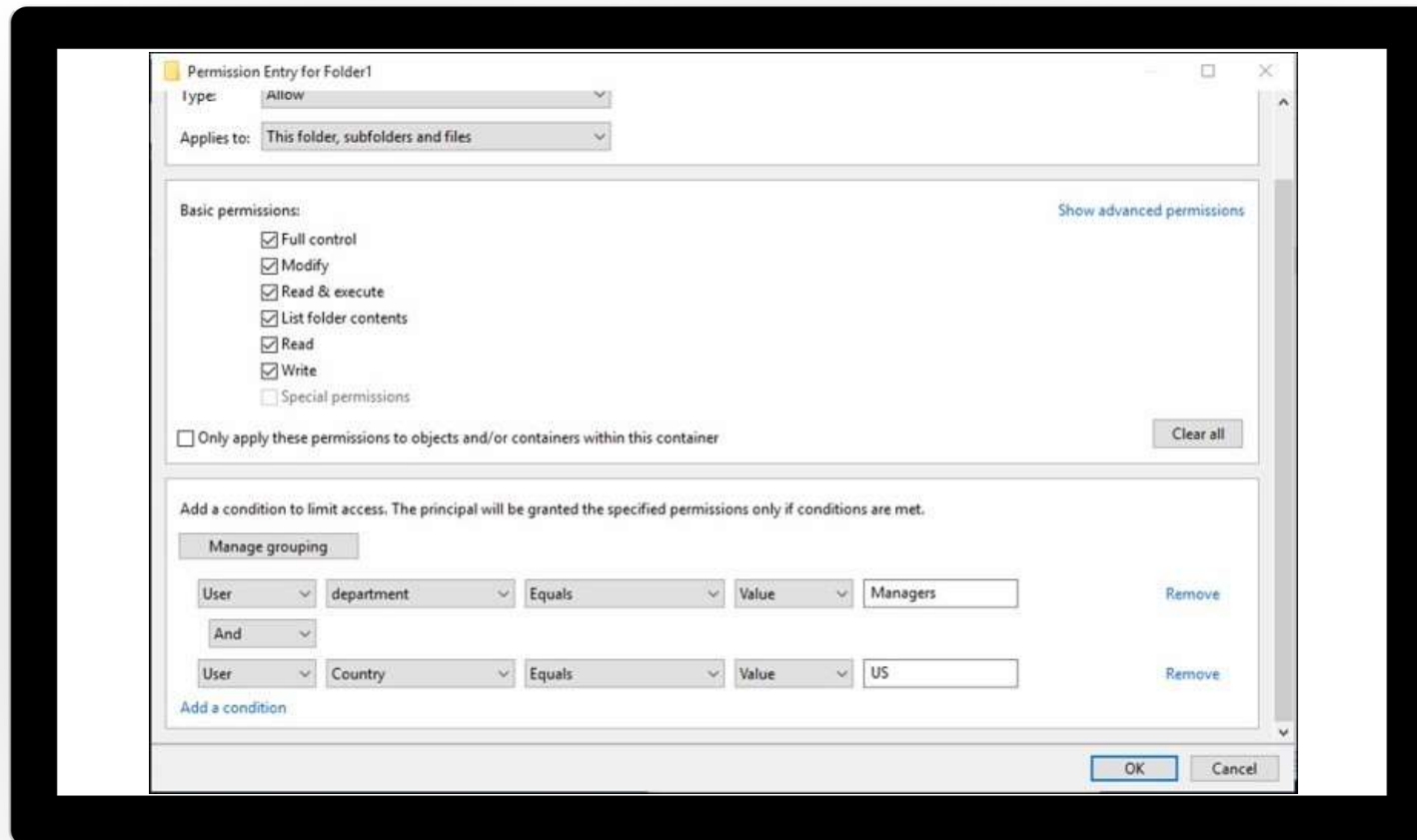
- Control access on NTFS and ReFS file system
- Can be added for groups, users, and computers:
 - Cumulative for group members
- Can be assigned to:
 - Files
 - Folders
 - Volumes (root folder)
- Permissions can be:
 - Allow
 - Deny (takes precedence)
- Basic and advanced



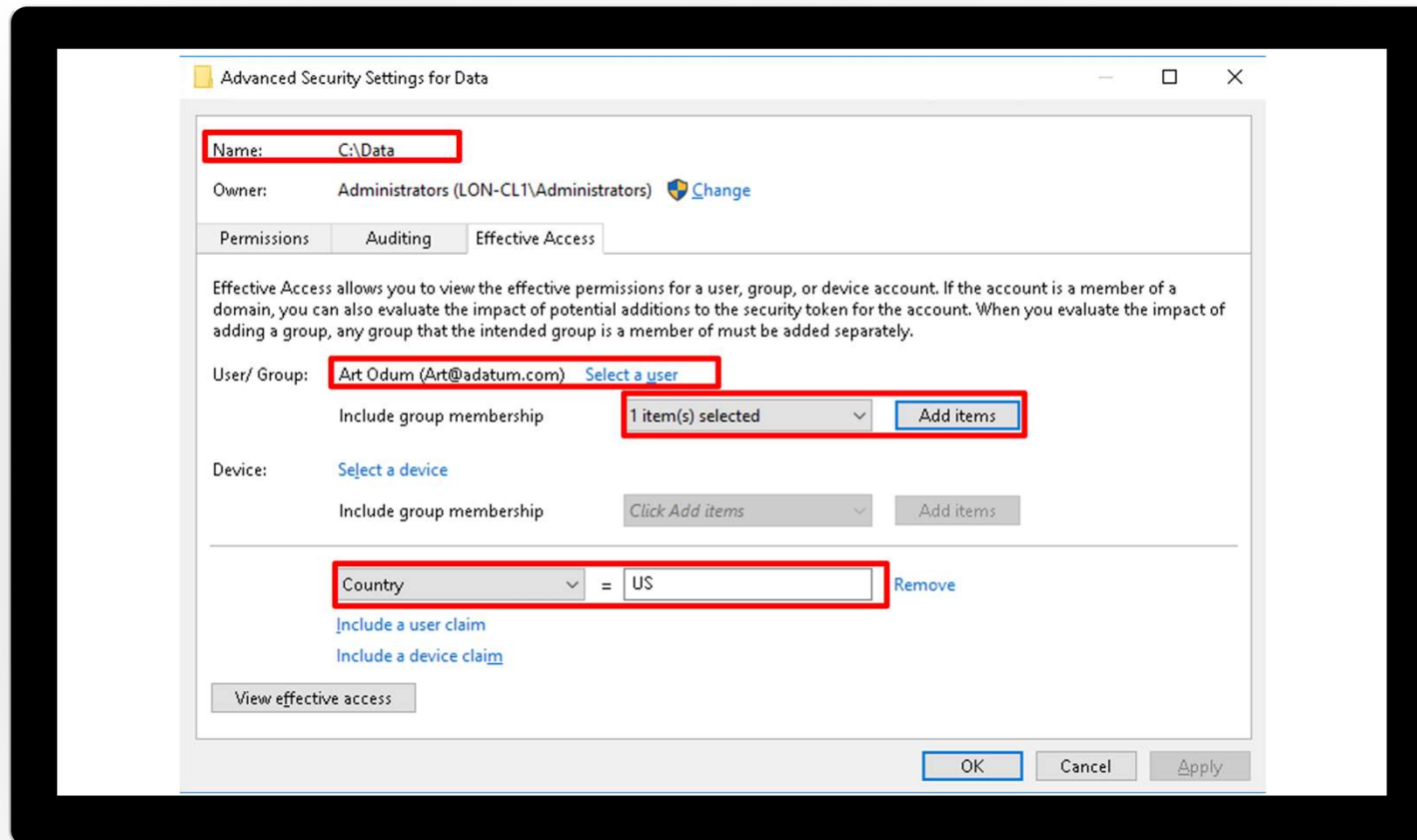
Understand permission inheritance



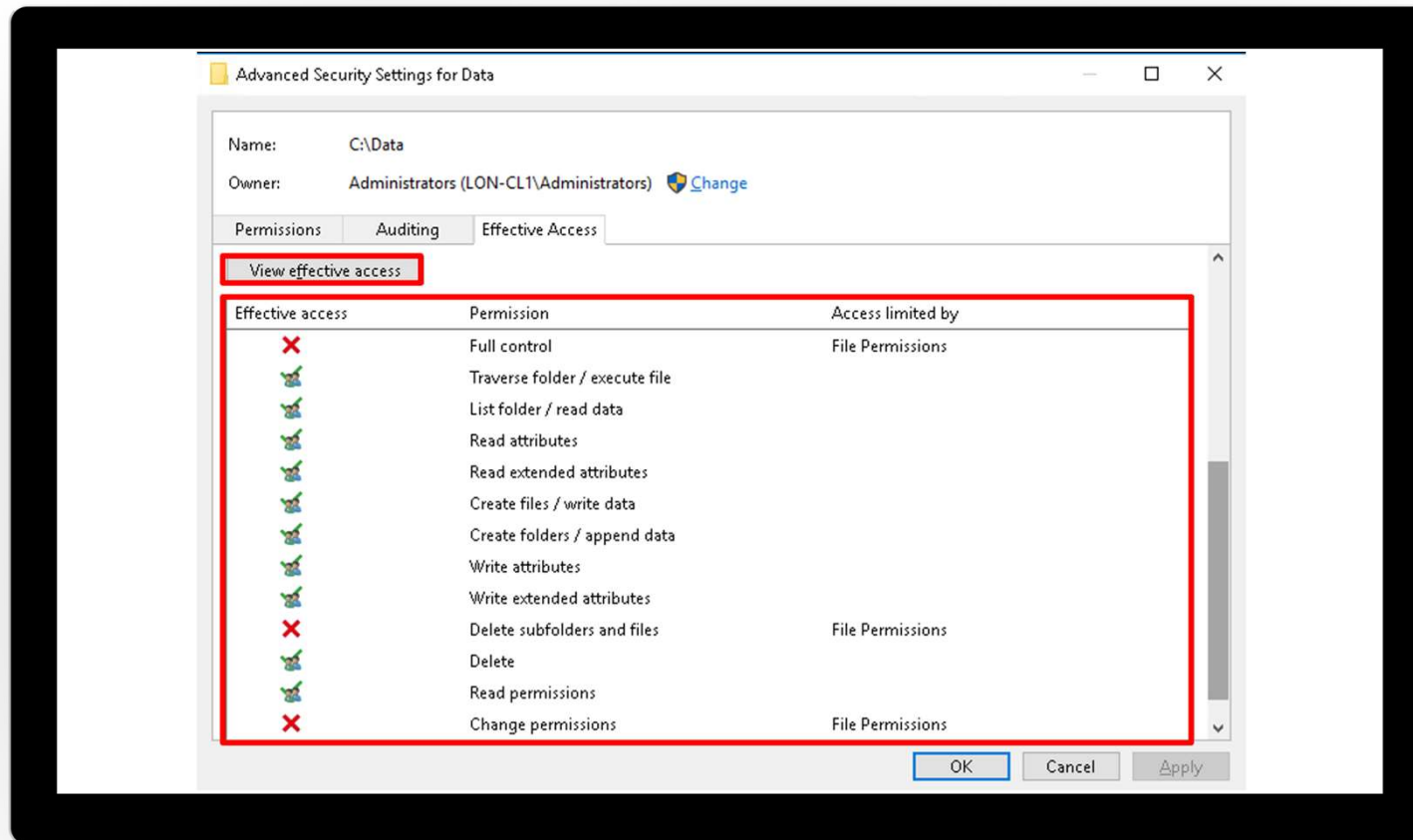
Implement conditions to limit file and folder access



Understand effective permissions

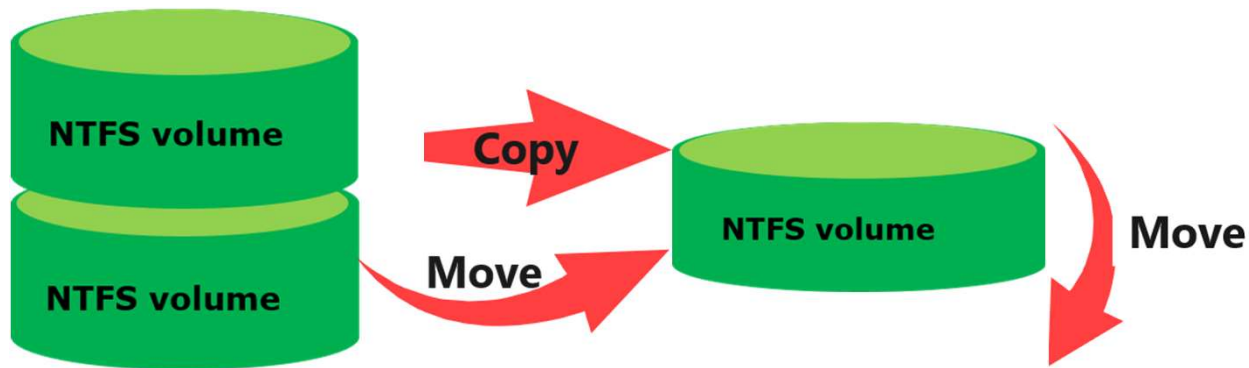


Use the effective access feature



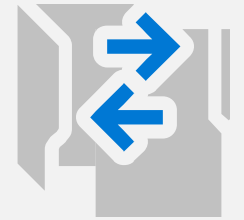
Examine permissions when copying and moving files

- When you copy files and folders, they always inherit the permissions of the destination folder
- When you move files and folders within the same volume, they keep their explicit permissions
- When you move files and folders to a different volume, they inherit the permissions of the destination folder



Knowledge Check

Lesson 3: Configure and manage shared folders



Lesson 3: Configure and manage shared folders

- Explore shared folders
- Examine methods available for connecting to share folders
- Examine shared folder properties
- Combine shared folder and NTFS permissions

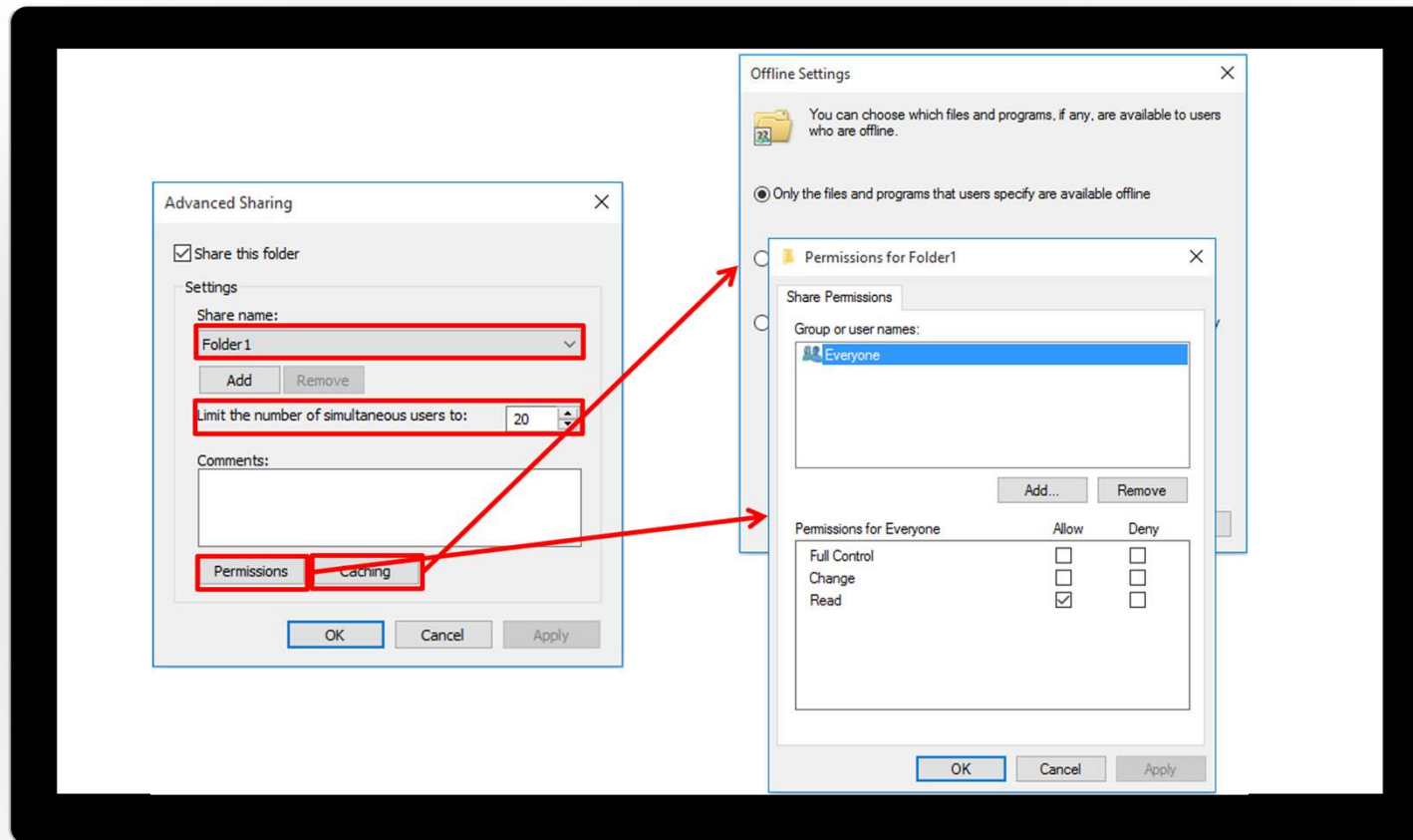
Explore shared folders

- Shared Folders are accessible over a network
- You can share only volumes and folders
- Share permissions control access over a network:
 - They do not apply if users access the content locally
- You must have Administrative permissions to create a share
- There are many ways to view shared folders:
 - File Explorer
 - The Shared Folders snap-in
 - **Net view\\computername /all**
 - The **Get-SmbShare** cmdlet
- You can access a shared folder by using UNC

Examine methods available for connecting to share folders

- Connecting to a shared folder
- Sharing folders by using Shared Folders snap-in
- Sharing folders using File Explorer
- Using the Share with option
- Advanced sharing
- Sharing folders using the command line
- Sharing folders using PowerShell

Examine shared folder properties



Combine shared folder and NTFS permissions

Permissions combine

- File/Folder permissions apply – but filter against shared permissions
- Most restrictive becomes effective permissions between the two

Shared Permission	File Permission	Effective Permission
Read	Full Control	Read
Change	Full Control	Modify
Full Control	Read	Read

Knowledge Check

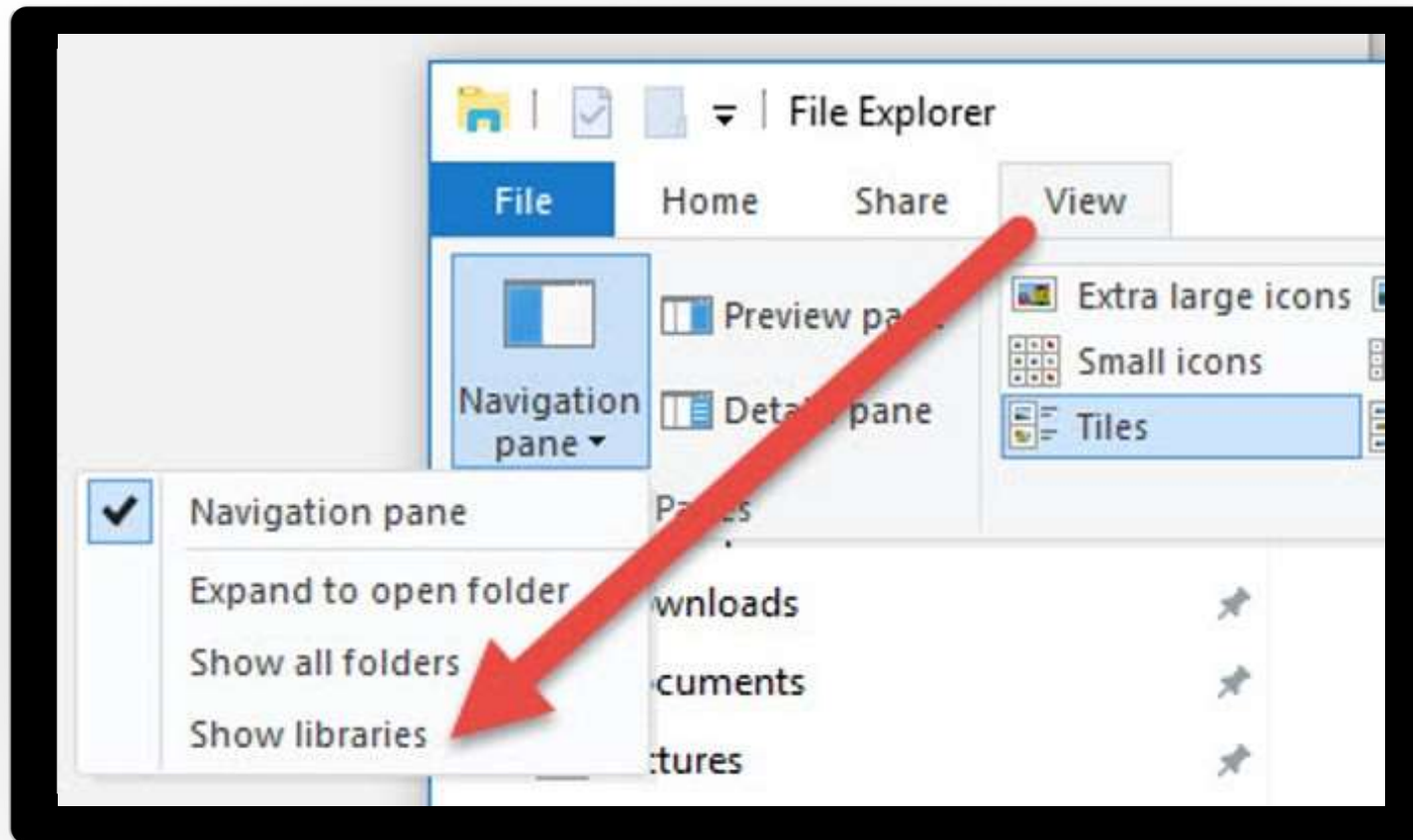
Lesson 4: Manage user files



Lesson 4: Manage user files

- Understand Windows Libraries
- Evaluate OneDrive
- Enable OnDrive
- Examine Work Folders

Understand Windows Libraries



Evaluate OneDrive

Microsoft provides cloud storage solutions for files.

- Free with Microsoft Account
 - Microsoft 365 subscribers get 1TB of storage
 - Business and Enterprise Microsoft 365 plans provide additional organizational features.
 - Both scenarios can co-exist on the same device
- Solves common challenges
 - Protects data loss against hardware failure
 - Simplifies migrations to new devices
 - Access to work/school data from anywhere
- Features:
 - Known Folder Move
 - OneDrive files on-demand
 - File restore, - Recycle bin
 - Auditing and reporting
 - Encryption of data in transit and (through BitLocker) at-rest.

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Enable OneDrive

- You must have a personal or work/school Microsoft account to sign in to OneDrive
- The OneDrive app for desktop provides options for file and folder sync
- You can restrict access to OneDrive by using Group Policy.

Examine Work Folders

- Allow users to access their individual data:
 - Users can access only their own Work Folders
- Enable data to be stored centrally on traditional file servers:
 - File servers must be running Windows Server 2012 R2 or newer
- Allow users to use multiple and various devices for access:
 - Regardless of whether devices are domain-joined
- Allow access from any location with the connectivity:
 - Local copy is available without network connectivity
- Ensure compliance with company policy:
 - Access control, quotas, file screening, classification, and more
 - Local data copy can be encrypted and wiped remotely

Knowledge Check

Practice Labs

Configuring and Managing Permissions and Shares



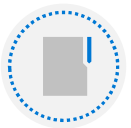
Using Conditions to Control Access and Effective Permissions



Troubleshooting File Access Issues



Synchronizing files with OneDrive



Module Review

